

# **A Regime Theory of Joy: Threshold Dynamics, Evaluative Monitoring, and Access Conditions**

*The “Ease” Regime as a Permissive Control Configuration*

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## Abstract

This paper proposes that intense positive affect (“joy”) is not produced by reward, learning, or goal-directed activity, but emerges from a distinct control regime, termed the *Ease regime*. Access to this regime is governed by a threshold variable ( $Z$ ), reflecting the degree of evaluative monitoring and optimization. When  $Z$  is high, cognition operates in an evaluation-dominated mode characterized by continuous correction, goal orientation, and performance tracking, constraining affective intensity. When  $Z$  falls below a critical threshold, evaluative processes fail to stabilize, allowing discrepancies to remain informational and enabling access to high-intensity positive affect.

Entry is abrupt, unambiguous, and not preceded by gradual improvement, consistent with a regime shift rather than a learning process. A central blocking mechanism,  $Z\_shift$ , is identified as a self-reinforcing loop in which attempts to recover the state induce persistent monitoring that prevents re-entry. The model predicts a measurement paradox: methods that recruit self-monitoring suppress the phenomenon they attempt to observe.

*The framework offers a falsifiable alternative to reward-based and training-based accounts, positioning joy as a threshold-dependent property of system dynamics rather than an outcome of optimization.*

## Introduction

Contemporary models of positive affect predominantly conceptualize joy as the outcome of reward processing, goal attainment, or affective valuation (Berridge & Kringelbach, 2015; Schultz, 2016). Within this framework, increases in positive affect are typically treated as continuous, scalable, and directly producible through identifiable inputs such as reinforcement, prediction error, or cognitive appraisal (Sutton & Barto, 2018; Rutledge et al., 2014). However, this view struggles to account for a class of phenomenological observations in which high-intensity positive states appear abruptly, resist voluntary induction, and collapse under attempts at control or stabilization, as reported in initial observational work (Morin, 2026).

These observations suggest that certain forms of positive affect are not well described as outputs of additive reward mechanisms, but instead reflect access to a distinct regime of cognitive-affective organization. In these cases, the transition into the state is discrete rather than gradual, often described as a sudden shift in the structure of experience, accompanied by a qualitative change in how discrepancies, goals, and sensory inputs are processed. Notably, these states cannot be reliably produced through effort, training, or optimization strategies, and are often disrupted by the very processes typically assumed to generate them.

A central difficulty in studying such states lies in a methodological constraint that has remained largely unaddressed. Standard approaches to affect measurement, including self-report scales, guided introspection, and task-based evaluation, implicitly introduce evaluative demands that alter the underlying cognitive regime (Schooler, 2002; Wilson & Schooler, 1991). These procedures require subjects to monitor, interpret, and report their internal states, thereby engaging control processes that may prevent the emergence of the very phenomena under investigation. As a result, the absence or rarity of such states in experimental settings may not reflect their true accessibility, but rather the conditions imposed by the measurement process itself.

This leads to what can be termed a measurement paradox in affect research: the more precisely a high-intensity positive state is targeted, measured, or operationalized, the less likely it is to occur. In this sense, current methodologies may systematically underestimate or mischaracterize regimes of positive affect that depend on the suspension of evaluative monitoring. This issue is not limited to extreme or rare experiences, but may extend to common developmental phenomena, such as the apparent accessibility of intense positive affect during childhood and its progressive attenuation under adult cognitive constraints.

In response to these limitations, the present paper proposes a regime-based account of joy. Rather than treating positive affect as a scalar output of reward systems, joy is conceptualized as emerging from a specific configuration of cognitive control in which discrepancy signals remain informational and are not immediately converted into corrective demands. This configuration defines a permissive regime, characterized by reduced evaluative stabilization and altered dynamics of prediction error processing (Friston, 2010; Clark, 2013). Within this framework, entry into joy is governed by threshold dynamics rather than continuous accumulation. The transition occurs when

evaluative control fails to stabilize discrepancies, allowing a shift in the functional role of prediction error from corrective signal to experiential driver. This model generates distinct empirical predictions, including abrupt onset, all-or-none phenomenology, sensitivity to contextual evaluation load, and incompatibility with sustained goal-directed optimization.

By reframing joy as a regime shift rather than a reward output, this account aims to resolve inconsistencies in the existing literature and to provide a basis for new experimental approaches that do not interfere with the phenomena they seek to measure.

## **Reframing “Ease” as a Computational Regime**

The term “Ease” is used here as a provisional label for a specific configuration of cognitive-affective dynamics. However, the core claim of this paper does not depend on the existence of a distinct or proprietary category of experience. Rather, the phenomenon is better understood as an instance of a broader class of regime shifts in cognitive control.

In this context, a regime is defined as a stable configuration of relationships between key functional components, including discrepancy detection, evaluative monitoring, and action selection. Different regimes are characterized not by the presence or absence of these components, but by how they are coupled. The proposed account focuses on a particular configuration in which discrepancy signals are not immediately routed into corrective or goal-directed processes.

Under standard conditions, discrepancies between expected and observed states trigger rapid evaluation and correction. This coupling supports efficient goal pursuit and environmental adaptation, but also imposes a continuous evaluative structure on experience. In contrast, the regime described here involves a temporary decoupling between discrepancy detection and corrective authority. Discrepancies remain active, but their functional role shifts: instead of initiating control processes, they persist as informational signals within the system.

This shift has several consequences. First, it alters the temporal dynamics of processing, as discrepancies are no longer rapidly resolved. Second, it changes the experiential role of prediction error, which can become a direct contributor to affective intensity rather than a transient signal to be minimized. Third, it reduces the dominance of goal-directed structure, allowing perception and action to unfold without continuous optimization pressure.

Importantly, this regime should not be interpreted as a deficit or failure of control in a pathological sense. The same underlying architecture remains intact, and the system retains the capacity to re-engage evaluative processes at any moment. Rather, the regime reflects a change in control policy, specifically a reduction in the gain or authority assigned to evaluative monitoring.

From this perspective, “Ease” does not denote a unique state, but a point within a space of possible control configurations. Similar regime shifts may occur under different conditions, including pharmacological modulation, developmental stages, or specific task structures. The present proposal isolates one such configuration and characterizes its dynamics, without claiming exclusivity or categorical novelty at the level of phenomenology.

This reframing has two advantages. First, it situates the phenomenon within existing computational and cognitive frameworks, making it compatible with models of predictive processing, control theory, and reinforcement learning. Second, it avoids treating the phenomenon as an exceptional or unanalyzable category, instead allowing it to be studied in terms of parameter changes, coupling structures, and transition dynamics. In this sense, the contribution of the present work is not the introduction of a new experiential label, but the identification of a specific regime within an already shared architecture, along with a set of conditions under which this regime becomes accessible.

## **Relation to Existing Frameworks**

The regime proposed here overlaps superficially with several well-established constructs in the study of attention, affect, and consciousness. However, these similarities are primarily phenomenological and do not extend to the underlying dynamics or access conditions. A comparative analysis helps clarify these distinctions.

### **Meditative absorption (e.g., jhāna states)**

Meditative absorption states are typically characterized by sustained attention, progressive stabilization, and the cultivation of specific attentional skills over extended training periods. Entry into such states is often gradual and depends on the active maintenance of attentional focus.

By contrast, the regime described here does not require training or sustained attentional control. Entry is abrupt and cannot be achieved through deliberate stabilization. In fact, attempts to maintain or reproduce the state through controlled attention tend to disrupt it. This suggests that, while both involve altered experiential structure, they rely on different control policies: one based on attentional stabilization, the other on a reduction of evaluative monitoring.

### **Flow states**

Flow is commonly defined as a state of optimal engagement during goal-directed activity, where skill and challenge are well matched. It involves high task focus, efficient performance, and a tight coupling between perception and action within a structured objective.

The present regime differs in that it is not tied to task optimization or performance. It can occur in the absence of goals and is often disrupted by the introduction of explicit objectives. Rather than reflecting efficient control within a task, it reflects a temporary suspension of optimization pressures altogether. As such, it is anti-instrumental in its access conditions, even if it may coexist with ongoing activity.

### **Reward-based and reinforcement learning accounts**

Standard accounts of positive affect link pleasure to reward prediction error, reinforcement signals, or dopaminergic valuation processes. Within these models, positive affect is typically treated as a function of outcomes, expectations, and their discrepancies.

While prediction error appears to play a role in the present regime, its functional status differs. Instead of being rapidly minimized through updating or action, discrepancies persist and contribute directly to experiential intensity. This implies a shift in how prediction error is handled, from a control signal to an experiential driver. The difference is therefore not in the presence of prediction error, but in its integration within the control architecture.

### **Nostalgia and memory reactivation**

Certain aspects of the regime, such as the re-emergence of childhood-like affective qualities or the vivid reactivation of past experiences, may resemble nostalgia. However, nostalgia is typically content-driven and linked to autobiographical memory retrieval.

In contrast, the present account suggests that memory reactivation is secondary to a change in regime. When evaluative constraints are reduced, previously encoded experiential configurations may reappear through mechanisms such as pattern completion. The key distinction is that the affective shift is not caused by the content itself, but by the conditions that allow such content to regain its original structure.

## **Comparative Summary: Access Conditions, Dynamics, and Failure Modes**

Across existing frameworks, differences can be systematically characterized along three dimensions: access conditions, internal dynamics, and failure modes.

In meditative absorption, access depends on sustained attentional training and progressive stabilization. The internal dynamics are characterized by increasing coherence and reduction of distraction through active control. Failure typically occurs through attentional drift or insufficient stabilization, leading to a gradual loss of the state.

In flow states, access requires a structured task with a balance between skill and challenge. The dynamics involve tightly coupled perception–action loops in the service of goal-directed performance, with high efficiency and minimal conflict. Failure arises when this balance is disrupted, either through excessive difficulty (leading to anxiety) or insufficient challenge (leading to boredom), as well as through interruptions that break task continuity.

In reward-based accounts, access to positive affect depends on favorable outcomes, reward prediction errors, or reinforcement signals. The dynamics are driven by expectation–outcome comparisons, with rapid updating and integration into action policies. Failure occurs when rewards are absent, predicted, or devalued, leading to attenuation of affective response.

In nostalgia and memory-based accounts, access is content-dependent, typically triggered by cues associated with autobiographical memory. The dynamics involve retrieval and reconstruction of past experiences, often accompanied by affective coloration. Failure arises when cues are absent, overused, or lose associative strength, reducing the intensity of reactivation.

By contrast, the regime proposed here is defined by access conditions that are anti-instrumental and highly sensitive to evaluative load. Entry occurs when evaluative monitoring fails to stabilize discrepancies, rather than through training, task optimization, or specific content. The dynamics are characterized by a persistence of discrepancy signals in an informational mode, without rapid conversion into corrective demands. This allows prediction error to contribute directly to experiential intensity. Failure is immediate and systematic under reintroduction of evaluation, goal-setting, or attempts at control, which re-establish the standard coupling between discrepancy and correction.

This comparison highlights that the proposed regime is not distinguished by its phenomenological content alone, but by a specific configuration of control parameters and transition conditions. As such, it occupies a distinct position within the broader space of cognitive-affective regimes, defined less by what is experienced than by how the system processes discrepancy, evaluation, and action.

# **From Prediction Error to Embodied Affect: Signal Transformation in the Permissive Regime**

## **Modulation by Sensory Structure**

Within the permissive regime, certain classes of sensory input appear to modulate the intensity of positive affect without being sufficient to induce the state.

In particular, visual features associated with high salience, such as brightness, contrast, or properties resembling rarity cues (e.g., glow-like or “treasure-like” signals), are frequently reported to amplify subjective intensity. These features may act by increasing the persistence or propagation of discrepancy signals under reduced evaluative coupling.

A second, partially distinct mode of positive affect is also observed. This mode is typically associated with slow-tempo, low-frequency auditory input, often involving simple or minimally structured melodic patterns. It is described as less localized and less sharply defined than the primary mode, and may involve diffuse, low-contrast perceptual qualities.

Importantly, these inputs do not reliably produce the regime in isolation. Their effects appear contingent on prior access to the regime and should therefore be interpreted as modulatory rather than generative.

This distinction suggests that different input classes may interact with the regime through partially dissociable pathways, potentially corresponding to different modes of discrepancy propagation.



## **This is not nostalgia: ease as reactivation of encoded affective states**

Reactivation of encoded affective configurations.

The permissive regime may enable the reactivation of previously encoded experiential configurations associated with early-life positive affect. This reactivation is not equivalent to standard autobiographical recall, but reflects a reinstantiation of the original affective and perceptual structure of the experience.

For example, exposure to a previously encountered object (e.g., a childhood-related stimulus) may be accompanied by a rapid and coherent re-emergence of its associated affective tone. This reactivation appears to involve the joint reinstatement of sensory, affective, and bodily components, rather than the retrieval of abstract or narrative memory content.

Importantly, this process does not depend on voluntary recall and cannot be reliably induced through deliberate effort. In some cases, it may occur automatically and resist suppression, suggesting that it reflects a system-level change in access conditions rather than a controlled retrieval process.

From a mechanistic perspective, this phenomenon is consistent with processes resembling hippocampal pattern completion, in which partial cues trigger the reactivation of previously encoded configurations. Under standard evaluative conditions, such reactivation may be attenuated or fragmented by rapid categorization and control processes. Under reduced evaluative coupling, however, these configurations may re-emerge as coherent experiential states.

In this sense, the permissive regime does not generate new affective content, but restores access to configurations that remain encoded but are typically inaccessible under standard control dynamics.

### **Linking Reactivation to Regime Dynamics**

The reactivation of encoded affective configurations can be interpreted within the regime framework as a consequence of altered discrepancy dynamics under reduced evaluative coupling.

Under standard conditions, partial cues associated with past experiences may activate corresponding memory traces. However, these activations are typically short-lived or fragmented, as discrepancy signals are rapidly integrated into evaluative and corrective processes. This results in categorization (“this is a marble”), narrative framing, or goal-related processing, which prevents the full reinstantiation of the original experiential configuration.

Within the permissive regime, the reduction in evaluative gain alters this process. As the coupling between discrepancy detection and correction weakens ( $g < g^*$ ), activated signals are no longer rapidly resolved. Instead, they persist and propagate across associated representations.

In formal terms, partial cues generate a discrepancy signal  $\delta(t)$  that is not immediately damped, allowing an increase in effective discrepancy over time:

$$\delta_{\text{eff}}(t) = \int_0^t \delta(\tau) \cdot w(\tau) d\tau$$

This sustained signal can facilitate the coordinated reactivation of distributed components of the original encoding, including sensory, affective, and bodily elements. The result is not a reconstruction assembled through recall, but a re-emergence of the configuration as a coherent whole.

Within this framework, the intensity of the reactivated state depends on the persistence and integration of discrepancy signals. When evaluative gain remains low,  $\delta_{\text{eff}}(t)$  can reach levels sufficient to produce high-intensity positive affect. When evaluative processes re-engage, discrepancy signals are again rapidly resolved, fragmenting or terminating the reactivation.

This account predicts that memory-based reactivation should be highly sensitive to evaluative load. Under high evaluative conditions, cues should primarily elicit recognition or narrative recall. Under low evaluative conditions, the same cues may trigger full experiential reinstatement.

More generally, this mechanism suggests that the distinction between memory and perception becomes partially blurred within the permissive regime. Previously encoded configurations are not merely accessed as representations, but re-enter the system as active experiential states, driven by the same discrepancy dynamics that govern ongoing perception

## **All-or-None Entry and Regime Shift Dynamics**

Entry into the permissive regime is characterized by abrupt, unambiguous transitions rather than gradual changes. Reports consistently describe a clear onset, with little to no intermediate or uncertain states. Ambiguous or “partial” entries are rare, suggesting that access is governed by threshold dynamics rather than continuous accumulation.

The phenomenological profile associated with this transition is not used here as evidence for the underlying mechanism, but as a set of constraints that competing explanations must account for. Alternative interpretations, such as immersive distraction or gradual mood elevation, generate distinct and testable predictions.

The transition itself can be described as a regime shift, marked by a sudden and sustained increase in positive affect. This increase is not limited to hedonic tone but reflects a broader reorganization of experience. In some cases, the resulting intensity is reported as unusually high, occasionally approaching levels that are difficult to tolerate.

Importantly, the shift is multidimensional. Alongside elevated positive affect, reports frequently include increased sensitivity to reward, enhanced emotional variability, vivid imagery, spontaneous laughter, and a general amplification of experiential salience. The state is typically described as self-sufficient and non-teleological, emerging independently of explicit goals or outcomes.

A notable feature of this transition is its immediate recognizability. The contrast with baseline experience is often sufficiently pronounced that individuals report a clear sense of entering a qualitatively distinct mode of operation, rather than experiencing a simple increase along a familiar dimension. In some cases, the initial phase may include atypical or difficult-to-classify affective qualities, reflecting the abrupt reconfiguration of control dynamics. This pattern has been repeatedly observed in informal and exploratory settings (Morin, 2026).

### **Distinguishing Threshold Dynamics from Learning**

The regime account generates predictions that differ sharply from learning-based models.

Learning-based frameworks predict gradual improvement, correlation with performance or skill acquisition, and increasing reproducibility with repetition. By contrast, the present model predicts discontinuous transitions, weak or absent relationships with skill, limited reproducibility, and sensitivity to contextual and evaluative factors.

In this sense, repeated failure followed by abrupt success is not treated as noise, but as a signature of threshold-governed access. The absence of a learning curve is therefore a central prediction

rather than a limitation.

## **Joy as a Threshold-Dependent Regime Property**

In the present framework, joy is not treated as a direct product of inputs, actions, or optimization processes. Instead, it is conceptualized as a regime-dependent property of the system, contingent on the level of evaluative control.

This control is captured by a variable  $Z$ , which reflects the degree to which the system engages in continuous comparison, evaluation, and correction.  $Z$  indexes the strength of evaluative coupling that transforms discrepancies into goal-directed adjustments.

The system is governed by a critical threshold,  $Z\_threshold$ . When  $Z$  remains above this threshold, the system operates in an evaluation-dominated regime. When  $Z$  falls below this threshold, a qualitative shift occurs, and the system becomes permissive to a distinct regime in which high-intensity positive affect becomes accessible.

Formally:

- If  $Z > Z\_threshold$ , the system remains in an evaluation-dominated regime
- If  $Z < Z\_threshold$ , the system transitions into a permissive regime

This transition reflects a discontinuous change in system dynamics rather than a gradual modulation.

In the evaluation-dominated regime ( $Z$  high), processing is characterized by continuous monitoring, goal-directed behavior, performance tracking, and rapid error correction. Experience is structured by narrative integration and outcome relevance, and positive affect tends to be limited, unstable, and dependent on external contingencies.

By contrast, in the permissive regime ( $Z$  low), evaluative processes fail to stabilize discrepancies. This results in a relative suspension of optimization, reduced coupling between action and instrumental outcomes, and increased sensitivity to structured input. Within this configuration, high-intensity positive affect becomes accessible as a property of the system's dynamics rather than as the result of reward accumulation.

## **.Anti-Instrumental Access and Control Sensitivity**

Within the proposed framework, attempts to intentionally use, optimize, or reproduce the state are predicted to increase evaluative load and thereby reduce the probability of regime access. This follows from the assumption that goal-directed control and monitoring processes increase  $Z$ , pushing the system above the critical threshold.

Accordingly, access to the permissive regime is expected to be disrupted by conditions that introduce explicit or implicit optimization demands. These include goal setting, repetition with performance intent, and real-time evaluation or measurement. Rather than facilitating access, such processes are predicted to destabilize or prevent it.

This property distinguishes the present account from frameworks in which positive affect can be increased through training, optimization, or reward maximization. In particular, models grounded in reinforcement learning, performance-based flow, or training-dependent meditative practices assume that improved control or optimization enhances access to desirable states. By contrast, the current model predicts that increasing control in this sense reduces access by strengthening evaluative coupling.

From an intervention perspective, this implies that effective manipulations should target the structure of control rather than affect directly. Specifically, interventions that reduce monitoring, interrupt optimization processes, or limit narrative capture may increase the probability of regime entry. Importantly, the relevant constraint is not limited to explicit goals, but extends to continuous micro-level corrections applied to ongoing experience.

Within the permissive regime, external stimuli appear to modulate the intensity of the state without generating it. Inputs that introduce structured but manageable discrepancies, for example small and frequent prediction errors without strong demands for resolution, are predicted to amplify experiential intensity.

One class of stimuli consistent with this profile includes perceptual sequences characterized by high variability and low requirements for causal coherence, such as rapidly changing audiovisual content. These inputs may sustain discrepancy signals without triggering large-scale reorganization or corrective demands.

A key implication is that the challenge in sustaining high-intensity positive affect does not lie in identifying appropriate activities, but in preventing those activities from becoming instrumentalized. Once an activity becomes associated with goals, expectations, or evaluation, it is predicted to increase  $Z$  and reduce access.

Finally, the modulation of intensity by stimuli is contingent on prior regime access. In the absence of entry into the permissive regime, similar inputs are not expected to produce comparable effects, reinforcing the distinction between modulatory and generative factors. In this sense, the regime is not difficult to access because appropriate stimuli are rare, but because the conditions required for access are systematically undermined by optimization.

## **The Measurement Paradox in Affect Research**

The present framework implies a methodological constraint: when access to a regime depends on reduced evaluative engagement at entry, measurement procedures that recruit self-monitoring may suppress the phenomenon they aim to observe.

Common experimental practices, including real-time ratings, introspective prompts, and performance framing, require participants to monitor and report their internal states. These operations introduce evaluative load and are therefore predicted to increase  $Z_{\text{ctx}}$ , the contextual component of evaluative coupling. As a result, such procedures may reduce the probability of regime entry, producing false negatives or systematically attenuated observations.

In this sense, measurement is not neutral but acts as an intervention on the control dynamics of the system. The phenomenon may behave as a target that becomes less accessible under explicit detection, leading to a structural bias in standard experimental designs.

More generally, any manipulation that increases perceived evaluation, including observation, reporting requirements, or explicit instructions, is predicted to interfere with regime access. This suggests that the absence of high-intensity positive states in laboratory settings may reflect methodological constraints rather than underlying inaccessibility.

To address this issue, the model generates several methodological predictions. First, post-hoc measurement, in which reporting occurs after the experience, should increase the observed probability of regime entry relative to real-time assessment. Second, behavioral or indirect proxies may allow detection of regime transitions without requiring explicit evaluation. Third, experimental designs that minimize anticipatory monitoring, including partial blinding to the hypothesis or task purpose, should increase access probability.

An additional implication concerns spontaneous reporting. In conditions where evaluative demands are low, individuals may report the onset of the state without prompting. Such reports can serve as behavioral markers of regime transition, providing an alternative to direct measurement during the state itself.

Taken together, these considerations suggest that inconsistencies in the empirical literature on positive affect may, in part, reflect measurement-induced regime collapse. The present framework therefore calls for methodological approaches that account for the interaction between observation and control dynamics. This paradox implies that the most direct methods of measurement may be the least valid for this class of phenomena.

### **Nonlinear Dynamics and Multi-Scale Constraints**

The proposed system is not continuous in its access dynamics. Rather, access to the permissive regime is governed by a threshold constraint on the evaluative variable  $Z$ . When  $Z$  remains above a critical threshold, access is effectively prevented. When  $Z$  falls below this threshold, access becomes possible.

This transition is nonlinear and better described in terms of phase transitions or bifurcation dynamics than gradual accumulation. Positive affect, in this framework, does not scale continuously with inputs or rewards, but depends on whether the system crosses a critical boundary in its control parameters. As a result, regime accessibility is probabilistic and sensitive to small variations near the threshold.

$Z$  itself is not a unitary variable but reflects the combined influence of structural, contextual, and historical factors. When  $Z$  is high, monitoring processes remain stable and continuously convert discrepancies into corrective actions, preventing the emergence of the permissive regime. When  $Z$  is sufficiently reduced, evaluative processes fail to stabilize discrepancies, allowing the system to cross the threshold and transition into the permissive regime.

Once this transition occurs, monitoring may re-enter at any moment, producing a collapse of the regime. However, the dynamics are asymmetric: entry appears more constrained than persistence. In this sense, the system is entry-limited rather than stability-limited, consistent with hysteresis-like behavior.

To account for the multi-scale nature of  $Z$ , the model distinguishes three interacting components.  $Z_{acc}$  captures slow, accumulated influences, such as long-term exposure to evaluation, performance pressure, and goal enforcement.  $Z_{shift}$  refers to a structural loop in which attempts to

detect or recover the state introduce persistent anticipatory monitoring, effectively maintaining the system above threshold.  $Z_{ctx}$  captures transient contextual factors, such as perceived evaluation, time pressure, or task framing.

Formally, regime accessibility and collapse can be treated as a function of these interacting components:

$$Z = Z_{acc} + Z_{shift} + Z_{ctx}$$

and:

$$\text{Regime collapse} = f(Z_{acc}, Z_{shift}, Z_{ctx})$$

This multi-layer structure implies that access depends on the stacking of constraints across

timescales. Under high  $Z$ , discrepancies are rapidly resolved through categorization, evaluation, and goal-directed processing. Under low  $Z$ , discrepancies remain temporarily unresolved, allowing them to propagate and contribute to the emergence of high-intensity, non-instrumental positive affect.

This structure predicts that small contextual changes may have disproportionate effects near the threshold, while large interventions may have minimal effects when the system remains far from it.

## **Monitoring, Evaluation, and Optimization**

A central distinction in the present framework concerns the relationship between monitoring, evaluation, and optimization. Monitoring is defined here as a control-layer mechanism that continuously tracks discrepancies between expected and observed states. Optimization, by contrast, refers to a behavioral regime that emerges from sustained monitoring, in which discrepancies are systematically converted into corrective or goal-directed adjustments.

In this sense, optimization is not treated as a voluntary strategy, but as a mode of operation induced by persistent monitoring. When monitoring is continuously active, discrepancies are assigned corrective relevance and routed toward action preparation, even in the absence of explicit goals. Evaluation can be understood as an observable manifestation of this process, rather than its underlying mechanism.



Within this framework, monitoring functions as a stabilizing layer that maintains the system in an evaluation-dominated regime. Its activity is continuous and self-reinforcing, as the conversion of discrepancies into meaningful or goal-relevant signals sustains further monitoring. As a result, the system tends toward a default mode in which experience is continuously interpreted, categorized, and adjusted.

Importantly, the proposed control structure may not be directly measurable without altering the phenomenon itself. Direct introspection or reporting engages evaluative processes and therefore modifies the underlying dynamics. Accordingly, the empirical target is not monitoring per se, but its behavioral signatures. These include abrupt regime entry, sensitivity to evaluative framing, disruption by repetition or goal-setting, and dissociation between access and skill acquisition.

The framework also predicts that monitoring may be indirectly reinforced through its association with reward-related signals. States characterized by high evaluative engagement may be experienced as meaningful, competent, or task-relevant, thereby stabilizing the optimization regime. Conversely, states with reduced evaluative structure may be perceived as less legitimate or less aligned with task demands, further biasing the system toward reinstating monitoring.

Taken together, these dynamics suggest that the optimization regime is self-stabilizing, both through control mechanisms and through its alignment with reward and meaning attribution. As a consequence, experiential modes that do not rely on evaluation may be underrepresented or actively suppressed within standard cognitive operation. In this sense, evaluation does not merely describe experience, it actively reshapes the regime in which experience unfolds.

## **Regime Collapse, Hysteresis, and the Z\_shift Mechanism**

Within the proposed framework, collapse of the permissive regime corresponds to the re-engagement of a control layer that reassigns evaluative relevance to ongoing experience. This reintroduction of evaluative arbitration restores the coupling between discrepancy detection and corrective processes, thereby terminating the regime.

Empirically, collapse is often associated with the emergence of meta-evaluative processes, such as monitoring the quality, duration, or meaning of the state. These processes are predicted to be sufficient to alter the control dynamics of the system, even in the absence of changes in external input or reward structure.

A key property of the system is its asymmetric stability. Entry into the regime appears fragile and constrained by threshold conditions, whereas persistence, once established, can tolerate partial re-engagement of evaluative processes. This asymmetry is consistent with hysteresis-like dynamics, in which the authority of evaluative control depends on the recent state history of the system.

Formally, entry requires the suppression of evaluative gain below a higher threshold, while persistence allows for its partial return without immediate collapse. Collapse, however, reinstates evaluative authority in a way that prevents rapid re-entry, even under conditions that were previously sufficient for access.

A central component of this dynamic is *Z\_shift*, defined as a structural loop in which attempts to detect, monitor, or recover the state introduce sustained anticipatory evaluation. In this loop, the intention to access the state leads to checking, and checking leads to persistent monitoring, maintaining the system above threshold. Over time, this loop can become automatic, transforming occasional evaluation into a stable background process.

This mechanism suggests a critical transition in which the system shifts from intermittent monitoring to continuous anticipatory evaluation. One plausible developmental implication is that this transition may occur during adolescence, not as a gradual change, but as a discrete shift linked to the realization that the state does not reliably return when expected. This realization may trigger the emergence of anticipatory testing, stabilizing the *Z\_shift* loop and reducing spontaneous access to the permissive regime.

## **Developmental Transition and Quiet Affect**

Within the present framework, developmental differences in positive affect are not interpreted as changes in capacity, but as transitions between control regimes. Children are not assumed to possess a greater intrinsic ability for positive affect. Rather, they are less likely to operate under a stable optimization regime.

In early development, discrepancies are more likely to remain informational rather than being systematically converted into corrective signals. Errors do not consistently trigger evaluation or negative tagging, and inefficiency is tolerated without immediate pressure for optimization. This allows for the persistence of open dynamics in which perception and action are not continuously structured by goal-directed correction. Play may contribute to stabilizing these dynamics by preventing premature closure and limiting the imposition of evaluative constraints.

In terms of the present model, this corresponds to relatively low accumulated evaluative load (*Z\_acc*), the absence or weak formation of anticipatory monitoring loops (*Z\_shift*), and low contextual evaluative pressure (*Z\_ctx*). Under these conditions, access to the permissive regime is more frequent, making it a plausible baseline mode of operation in early development.

A further implication concerns the phenomenological reporting of such states. Children often do not explicitly report or categorize these experiences, not because they are absent, but because they do not become objects of evaluation or reflection. This can be described as a form of “quiet affect,” in which affective states are present but not labeled, narrated, or amplified.

This leads to a form of double masking in the apparent loss of such states over development. First, access to the regime decreases as evaluative control becomes more stable. Second, the original experiences were not consistently encoded as explicit, reportable events. As a result, their disappearance is not experienced as a loss, and may remain cognitively untraceable due to the absence of both memory labeling and conceptual representation.

### **Affective Quietude, Development, and the Failure of Nostalgia**

The present framework suggests that the apparent absence of explicit reports of high-intensity positive affect in childhood does not reflect rarity or concealment, but a difference in representational structure. When a regime is frequently accessible, it may not be registered as exceptional and therefore does not become an object of explicit attention or conceptualization. In this sense, the system may maintain the regime behaviorally without representing it explicitly.

A related implication concerns the relationship between engagement and evaluative preference. Standard accounts typically assume that engagement is accompanied by explicit liking, preference, or endorsement. By contrast, the present model predicts that engagement can occur in the absence of such markers. Individuals may remain highly engaged with stimuli without labeling the experience as “liked,” without reporting it, and without forming stable preferences.

This dissociation is particularly relevant in early development. When the system is still constructing its internal model of the environment, stimuli function less as confirmations of established categories and more as openings within model space. Under these conditions, prediction error may persist without immediate resolution, allowing perception to remain exploratory and structurally transformative rather than rapidly stabilized.

As evaluative and explanatory processes become more dominant, perception shifts toward rapid categorization and closure. Surprise is attenuated, and repetition suppression increases, reducing the persistence of discrepancy signals. This transition constrains the system’s ability to sustain regimes in which discrepancies remain informational.

This perspective also provides an account of the limited effectiveness of nostalgia. Nostalgic processes typically rely on comparison and recognition, explicitly referencing prior experiences and evaluating the present against them. In doing so, they reinstate the evaluative and narrative processes that suppress the original regime. The result is a preservation of representational content without restoration of the underlying control dynamics.

Within this framework, childhood positive affect is not defined by increased emotional intensity or arousal, but by a regime of affective quietude, characterized by reduced internal evaluation, minimal narrative structuring, and low control pressure. The defining feature is not how much is felt, but how little needs to be regulated or interpreted.

## **Structured Prediction Error and Engagement in Cartoons**

Cartoons provide a useful example of stimulus structures that may interact with the permissiveregime. They are typically characterized by weak or absent resolution, minimal goal-directed structure, and low or inconsistent payoff contingencies.

Rather than relying primarily on classical humor mechanisms such as setup and punchline, cartoons often exhibit dense sequences of perceptual and causal violations. Physical laws may be intermittently broken, proportions destabilized, and objects transformed without warning. These features generate sustained prediction error and perceptual instability.

From this perspective, what is commonly interpreted as humor may partly reflect an evaluative overlay imposed on more fundamental processes. An alternative account is that such stimuli operate through structured prediction error, maintaining discrepancy signals without requiring immediate resolution.

Within the present framework, children's engagement with cartoons may reflect access to a regime in which discrepancies remain informational rather than being rapidly converted into corrective or evaluative processes. Importantly, the absence of explicit reports of strong positive affect in such contexts does not imply its absence. Instead, it is consistent with a regime in which affect is not amplified, categorized, or reported.

More generally, these observations support the claim that prediction error does not require resolution to sustain engagement. Rather, sustained engagement may depend on the absence of enforced closure.

→ **Z modulates the probability that prediction error is closed vs propagated.**

## **Control Reduction, Signal Coherence, and Pharmacological Dissociations**

Pharmacological interventions that reduce constraint or evaluative control do not reliably produce the regime described here. Although such interventions may increase affective flexibility or reduce negative affect, they often fail to induce a coherent and stable transition into a high-intensity positive regime.

For example, ketamine, which has been associated with reduced activity in anterior cingulate and related control networks, as well as rapid antidepressant effects, does not consistently produce a sustained or structured increase in positive affect. Instead, its effects are often characterized by dissociation, perceptual instability, or diffuse affective changes.

This dissociation suggests that a reduction in evaluative control, taken in isolation, is not sufficient for regime entry. The present framework predicts that access depends not only on lowering control-related gain, but on how this reduction is implemented. Specifically, entry requires a selective attenuation of evaluative micro-optimization within an otherwise intact perceptual and salience architecture.

When control reduction is global or accompanied by signal degradation, discrepancy signals may lose coherence and fail to propagate in a structured manner. Under such conditions, the system may exhibit reduced constraint without achieving the organized persistence of discrepancy required for high-intensity positive affect.

This distinction allows the model to accommodate mechanistic evidence linking control-related structures to affective regulation, while avoiding reduction to a simple inhibition account. In particular, it predicts that interventions which preserve signal coherence while reducing evaluative coupling should be more effective in enabling regime transitions than those that broadly disrupt neural processing.

More generally, this perspective suggests that the relationship between control and affect is not monotonic. Reducing control can produce qualitatively different outcomes depending on whether underlying signal structure is preserved or degraded.

An additional implication concerns tolerance. Rather than reflecting only reduced reward sensitivity, tolerance effects may partly arise from anticipatory control processes, in which repeated exposure leads to increased monitoring and prediction of the intervention's effects, thereby reinstating evaluative coupling and reducing regime accessibility.

## **The Morin Z-Reduction Task (M-ZRT): A Discriminative Protocol**

The Morin Z-Reduction Task (M-ZRT) is not intended as a standard induction technique, but as a discriminative protocol designed to differentiate between two classes of models: gradual, learning-based accounts and threshold-based regime-shift accounts.

In learning-based frameworks, repeated exposure to a task is expected to produce gradual improvement, increased reproducibility, and performance-dependent effects. By contrast, the present model predicts a distinct signature: repeated failures, followed by abrupt onset, a qualitative transition, and persistence beyond the initial context. The presence of discontinuous success following unsuccessful attempts is therefore treated not as noise, but as evidence of threshold-governed access.

Crucially, the M-ZRT is constructed to avoid the stabilization of evaluative control. Attempts to directly reduce evaluation are themselves predicted to increase monitoring, creating a self-reinforcing loop that prevents regime entry. For this reason, control over monitoring must be indirect. Any protocol that becomes stable, repeatable, or optimizable is expected to reinstate evaluative coupling and thereby reduce its own effectiveness.

Within this framework, the task is designed as a brief, non-instrumental perturbation of ongoing activity. Short sequences of actions may be more effective than single manipulations, as isolated interventions may be too weak or ambiguous, whereas extended sequences risk becoming structured and subject to optimization. The aim is to disrupt motor and cognitive continuity without allowing the formation of a repeatable routine.

A key constraint of the task is the absence of explicit goals, metrics, evaluation, repetition, or expectation. These conditions are intended to minimize evaluative load and prevent the reactivation of monitoring processes.

As an illustrative example, participants may engage in an ongoing activity (e.g., navigating a virtual environment) while introducing brief, non-instrumental perturbations, such as generating a transient mental image without attempting to maintain it, followed by an irregular, non-rhythmic motor action. The sequence is terminated without continuation or evaluation. The task is performed sparingly, and not repeated in a structured manner.

The model further predicts specific failure modes corresponding to the reintroduction of evaluative processes. These include anticipatory commitment, premature closure, performance monitoring, meaning attribution, corrective adjustment, and expectation-driven resolution. The reappearance of these processes is expected to block or terminate regime entry.

## **Perturbation Classes in the M-ZRT: Openness, Salience, and Suspension**

The M-ZRT can be decomposed into a set of minimal perturbation classes that differentially target evaluative control processes. These perturbations are not defined by their content, but by their functional effect on goal closure, attentional allocation, and motor continuity.

**Openness** refers to transient operations that weaken immediate goal closure or deliberate control. These perturbations introduce elements that are not used to advance task objectives and are not maintained or evaluated. For example, the generation of a brief, low-purpose mental image during ongoing activity may function as an openness perturbation, provided that it is neither sustained nor used instrumentally. Similarly, introducing an incomplete thought (e.g., “what if...”) without resolving or revisiting it disrupts closure without engaging problem-solving processes.

**Salience** refers to the temporary amplification of a single perceptual or internal element without converting it into a goal. These perturbations increase the local prominence of a signal while avoiding integration into task-relevant structure. For instance, the spontaneous fading of a mental image may act as a salience event by briefly capturing attention through a minimal internal change. Likewise, selecting a single sensory input (e.g., a sound) and momentarily treating it as dominant, without attaching relevance or action to it, produces a transient salience shift.

**Suspension** refers to brief interruptions of ongoing optimization or strategic continuity. These perturbations disrupt the flow of goal-directed behavior without introducing an alternative objective. For example, a short, non-instrumental interruption of movement followed by an uncorrected change in direction can break motor continuity while avoiding compensatory adjustment or explanation.

These three classes can be combined to produce composite perturbations that target evaluative coupling from multiple angles. In particular, sequences that introduce openness, followed by a salience event, and terminate with a suspension may be effective in disrupting monitoring without allowing the formation of a repeatable structure.

Importantly, the target configuration is not reduced activity or disengagement, but a state in which salience is preserved while evaluative control is attenuated. The aim is therefore not to suppress processing, but to decouple salience from optimization.

These perturbation classes are defined functionally and may be implemented across modalities, tasks, or environments.

## Discussion

The present framework proposes that high-intensity positive affect is not the outcome of reward accumulation, skill acquisition, or sustained practice, but the expression of a distinct control regime defined by the temporary failure of evaluative stabilization. This shifts the explanatory focus from content and stimulus properties to system-level dynamics governing access conditions. In this view, joy is not produced, but becomes accessible when monitoring and optimization processes fall below a critical threshold.

This perspective challenges dominant gradualist models. Learning-based accounts predict progressive improvement, reproducibility, and performance dependence. In contrast, the Ease framework predicts discontinuity, independence from skill, and abrupt transitions following repeated failure. The existence of such discontinuities, if empirically confirmed, would support a threshold-based model over incremental learning explanations.

A central contribution of the model is the identification of a structural blocking mechanism ( $Z_{\text{shift}}$ ), whereby attempts to access the state induce anticipatory verification that stabilizes into persistent monitoring. This loop provides a parsimonious explanation for the difficulty of voluntarily re-entering high-intensity positive states despite intact reward systems. Importantly, it suggests that the primary obstacle is not insufficient stimulation, but the presence of a self-sustaining control process.

The framework also introduces a methodological implication for affect research. If access depends on reduced evaluation at entry, then common experimental practices such as real-time reporting, introspective prompting, or performance framing may systematically suppress the phenomenon under investigation. This “measurement paradox” implies that some inconsistencies in the literature may reflect methodological interference rather than true absence of effect. Approaches relying on post-hoc reports, behavioral markers, or reduced hypothesis transparency may therefore be more appropriate for capturing regime transitions.

The account further offers an alternative interpretation of childhood affect. Rather than reflecting heightened emotional capacity, children may operate in a low-evaluative regime in which affect is not continuously monitored or categorized. This could explain both the apparent intensity of early-life experiences and the difficulty of recalling or reproducing them in adulthood. In this context, phenomena often described as nostalgia may fail not because the underlying representations are lost, but because the control regime required for their reactivation is no longer accessible.

At the stimulus level, the model predicts that environments rich in structured prediction error, such as cartoons or music videos, are more likely to modulate the intensity of the state once the regime is accessible, without being sufficient to produce it.

This distinction clarifies why similar stimuli can appear ineffective under high evaluative load yet become highly potent under permissive conditions.



Several limitations should be acknowledged. The construct of Z is currently operationalized at a theoretical level and requires empirical grounding through measurable proxies. Additionally, while the framework emphasizes control dynamics, its neural implementation remains underspecified. Candidate mechanisms may involve interactions between salience, monitoring, and control networks, but this remains to be tested. Finally, the phenomenological reports described here, while consistent, require systematic validation under controlled conditions.

Future work should focus on developing experimental paradigms capable of detecting discontinuous regime transitions without inducing evaluative interference. The proposed behavioral protocol (M-ZRT) provides a starting point for such investigations, particularly as a falsification tool against gradualist accounts. More broadly, this framework suggests that access to high-intensity positive affect may depend less on optimizing inputs and more on altering the conditions under which the system ceases to optimize.

## **Implications for Development, Psychopathology, and Learning**

If positive affect is partially governed by regime dynamics rather than solely by reward magnitude or valuation processes, this has broad implications across multiple domains of cognitive science.

### **Development**

The proposed framework offers a reinterpretation of developmental changes in affective experience. Early childhood is often characterized by frequent access to high-intensity positive states that appear difficult to reproduce in adulthood. Rather than attributing this difference to changes in reward sensitivity or novelty exposure alone, the present account suggests that developmental shifts in evaluative monitoring may play a central role.

Specifically, the progressive installation of stable evaluative and goal-directed control during adolescence may increase the baseline coupling between discrepancy detection and correction. This would reduce the probability of entering regimes in which discrepancies remain informational. From this perspective, the apparent “loss” of intense positive affect is not due to disappearance of capacity, but to a change in access conditions driven by control architecture.

### **Psychopathology**

The model also has implications for conditions involving altered affective regulation. Many forms of anxiety and mood disorders are associated with heightened evaluative monitoring, persistent error checking, and increased sensitivity to discrepancies. Within the present framework, such conditions can be understood as regimes in which the coupling between discrepancy and correction is excessively strong or rigid.

This predicts a reduced accessibility of permissive regimes of positive affect, not because reward systems are impaired per se, but because evaluative processes prevent the stabilization of such regimes. Conversely, certain pharmacological or behavioral interventions that reduce evaluative gain may transiently increase access to these states, although potentially at the cost of reduced control in other domains.

Importantly, this account distinguishes between reducing negative affect and enabling access to high-intensity positive regimes. These may involve overlapping but non-identical mechanisms, suggesting that standard anxiolytic or antidepressant approaches may not directly restore access to such states.

## **Reward and reinforcement learning**

The framework challenges the assumption that increasing reward or positive prediction error is sufficient to produce high-intensity positive affect. If prediction error is rapidly converted into corrective signals under standard control regimes, its contribution to subjective experience may be limited.

This suggests that the experiential impact of prediction error depends on its integration within the control architecture. Under reduced evaluative monitoring, prediction error may accumulate or persist in a way that enhances affective intensity. Under high evaluative gain, the same signals may be rapidly neutralized through updating and action.

As a result, interventions aimed at modulating reward signals alone may have limited effects unless they also alter the coupling between discrepancy detection and correction.

## **Learning and exploration**

Finally, the model has implications for learning and exploratory behavior. Standard accounts emphasize the role of prediction error minimization and reward maximization in guiding behavior. However, if certain regimes allow discrepancies to persist without immediate correction, this may create conditions for a different mode of exploration.

In such regimes, behavior may become less constrained by immediate optimization and more driven by local salience or unfolding perceptual dynamics. This could facilitate forms of exploration that are not directly tied to goal attainment, but still contribute to learning by exposing the system to a broader range of states and transitions.

This perspective suggests that alternating between regimes of tight control and regimes of reduced evaluative coupling may be beneficial for balancing exploitation and exploration. The present account isolates one such regime and provides a framework for studying its properties and transitions.

## **Predictions and Falsifiability**

The regime-based account proposed here generates a set of specific, testable predictions that distinguish it from reward-based, attentional, and memory-driven models of positive affect.

### **1. All-or-none transition dynamics**

If entry into the proposed regime is governed by threshold dynamics, then transitions into high-intensity positive affect should be abrupt rather than gradual. Reports of intermediate or continuously increasing states should be rare or unstable. *Test:*

Continuous self-report or behavioral proxies should show discontinuities rather than smooth increases. If affect reliably scales in a graded manner under controlled conditions, this would challenge the regime-shift account.

### **2. Negative effect of measurement on state occurrence**

If evaluative monitoring prevents regime entry, then standard measurement procedures should reduce the probability of observing the state.

*Test:*

Compare conditions with real-time self-report (ratings, prompts) versus post-hoc reporting. The model predicts a significantly lower incidence and intensity of the state under real-time evaluation. Failure to observe this difference would weaken the measurement paradox claim.

### **3. Anti-instrumental access conditions**

If the regime depends on reduced goal-directed control, then attempts to intentionally induce the state should decrease its probability of occurrence.

*Test:*

Instruct one group to “try to maximize positive affect” and another to perform a non-instrumental task without explicit goals. The model predicts lower incidence in the goal-directed group. If deliberate optimization reliably increases occurrence, the model is contradicted.

#### **4. Sensitivity to evaluative load (Z\_ctx)**

If evaluative load directly modulates access, then even subtle increases in perceived evaluation should suppress the state.

*Test:*

Manipulate context (e.g., being observed, performance framing, time pressure). The model predicts sharp reductions in occurrence under high evaluative conditions. If the state remains robust under such conditions, the model is weakened.

#### **5. Dissociation from reward magnitude**

If the regime is not driven by reward intensity, then increasing rewards should not reliably produce the state.

*Test:*

Provide high-reward versus low-reward conditions while keeping evaluative load constant. The model predicts weak or inconsistent effects of reward magnitude. Strong monotonic effects would favor reward-based accounts.

#### **6. Prediction error as experiential driver under reduced control**

If prediction error contributes directly to experience only under reduced evaluative coupling, then its affective impact should depend on control conditions.

*Test:*

Introduce controlled prediction errors under high vs. low evaluative load. The model predicts increased affective intensity only in low-evaluation conditions. If prediction error produces similar effects regardless of control context, the model is challenged.

#### **7. Immediate collapse under reintroduction of evaluation**

If the regime depends on suspended evaluation, then reintroducing evaluative processes should rapidly terminate the state.

*Test:*

Interrupt ongoing high-affect states with evaluative prompts (e.g., “rate your current feeling”). The model predicts immediate or near-immediate collapse. Persistence despite evaluation would contradict the model.

## **8. Developmental gradient linked to evaluative stabilization**

If developmental changes in affect are driven by increased evaluative coupling, then children should show higher spontaneous access independent of reward conditions.

*Test:*

Compare spontaneous high-affect episodes across age groups under matched reward and environmental conditions. The model predicts higher frequency in younger populations. Absence of such a gradient would weaken the developmental claim.

## **9. Limited trainability and poor reproducibility**

If the regime cannot be stabilized through practice, repeated attempts to induce it should not increase success rates and may decrease them.

*Test:*

Longitudinal attempts to train access using repeated protocols. The model predicts no reliable learning curve and possible decline due to increasing anticipation and monitoring. Strong improvements with training would contradict the model.

## **10. Context-dependent memory reactivation**

If memory reactivation is secondary to regime entry, then cues alone should be insufficient without the appropriate control conditions.

*Test:*

Present highly nostalgic cues under high vs. low evaluative load. The model predicts strong reactivation only in low-evaluation contexts. If cues reliably trigger the state regardless of context, the model is challenged.

## **Summary constraint**

A key feature of this framework is that multiple predictions converge on a single mechanism: the coupling strength between discrepancy detection and evaluative correction. Evidence that systematically dissociates state occurrence from this coupling, or demonstrates robust induction under high evaluative control, would directly falsify the model.

## **Strong Claims and High-Risk Predictions**

The present account entails a set of strong claims that depart significantly from standard views of positive affect. These claims are intentionally high-risk: each can be directly challenged by empirical data.

### **1. Positive affect cannot be directly optimized**

The model predicts that any systematic attempt to optimize, maximize, or stabilize high-intensity positive affect will reduce its probability of occurrence.

This is a strong inversion of standard reward-based assumptions. If participants can reliably increase the frequency or intensity of such states through explicit strategies, training protocols, or optimization procedures, the regime account is fundamentally undermined.

### **2. Measurement destroys the target phenomenon**

The model predicts that real-time measurement of high-intensity positive affect will systematically suppress or eliminate the state.

This goes beyond a mild observer effect. It implies that commonly used methods, such as continuous ratings or in-task introspective probes, do not merely bias results but may prevent the phenomenon from occurring at all.

If high-intensity positive states can be robustly elicited and maintained under continuous measurement, this claim is falsified.

### **3. Reward is neither necessary nor sufficient**

The model predicts that high-intensity positive affect can occur in the absence of significant reward, and conversely, that high reward does not reliably produce it.

This directly challenges reinforcement-based accounts. If increasing reward magnitude consistently and predictably increases the probability of entering such states, the regime hypothesis loses explanatory power.

### **4. Prediction error requires decoupling to become experiential**

The model predicts that prediction error contributes to subjective intensity only when evaluative correction is attenuated.

Under standard conditions, prediction error is rapidly neutralized through updating and action. Only when this process is disrupted does prediction error persist long enough to shape experience. If prediction error produces strong affective responses independently of control context, this claim is invalidated.

### **5. The state collapses under awareness of its own conditions**

The model predicts that recognizing, monitoring, or attempting to maintain the conditions that allow the state will terminate it.

This includes meta-awareness such as “this is working” or “I should preserve this.” The claim is that the system cannot simultaneously sustain the regime and represent it as an object of control. If individuals can maintain the state while explicitly tracking and stabilizing it, the model is contradicted.

### **6. Developmental loss reflects control installation, not capacity loss**

The model predicts that the apparent decline of high-intensity positive affect from childhood to adulthood is driven by increased evaluative control, not by reduced reward sensitivity or novelty.

If controlling for evaluative load does not restore access in adults, or if children do not show higher spontaneous access under matched conditions, this claim is weakened.

### **7. No reliable training curve exists**

The model predicts that repeated attempts to induce the state will not produce a stable learning curve and may even reduce success rates due to anticipation and monitoring.

This contradicts the majority of skill acquisition frameworks. If training leads to consistent, reproducible access, the regime model is invalidated.

### **8. Micro-context dominates macro-context**

The model predicts that small, often overlooked contextual factors (e.g., perceived evaluation, subtle goal framing, time pressure) will have larger effects on state occurrence than major variables such as reward magnitude or environmental richness.

If large-scale manipulations (e.g., immersive environments, high reward) consistently override micro-contextual factors, this claim fails.



## **9. The regime is inherently unstable under control**

The model predicts that the state cannot be stabilized without transforming into a different regime.

Attempts to prolong or fix the state will introduce control processes that alter its structure. Any stable, maintainable version would therefore no longer belong to the same regime. If long-duration, stable versions of the same phenomenological state can be maintained under control, the model must be revised.

## **10. A single parameter governs multiple phenomena**

The model predicts that a single underlying parameter, the coupling strength between discrepancy detection and evaluative correction, explains a wide range of observations across domains.

This includes state accessibility, developmental differences, measurement effects, and sensitivity to context. If these phenomena can be independently manipulated without affecting one another, the proposed unifying parameter is insufficient.

### **Interpretive constraint**

These claims are not incremental extensions of existing theories but structural alternatives. Their value lies in their vulnerability: each can be clearly supported or rejected. The framework therefore invites direct empirical confrontation rather than gradual integration.

## **Minimal Formalization of Regime Dynamics**

The proposed account can be expressed as a change in coupling between discrepancy detection and evaluative correction within a standard control architecture.

Let  $\delta(t)$  denote discrepancy or prediction error at time  $t$ . Under standard conditions, discrepancy is rapidly transformed into corrective signals through a gain-controlled process:

$$c(t) = g \cdot \delta(t)$$

where  $c(t)$  is the corrective drive and  $g$  is the evaluative gain parameter. High values of  $g$  correspond to strong, rapid conversion of discrepancies into goal-directed adjustments.

The key proposal is that access to the permissive regime depends on a reduction or temporary failure of this coupling. Specifically, when  $g$  falls below a critical threshold  $g^*$ , the system no longer enforces immediate correction:

$g < g^* \rightarrow$  decoupling regime

In this regime, discrepancies are not eliminated but persist as informational signals. Their temporal dynamics change accordingly. Instead of being rapidly damped, discrepancy signals accumulate or remain active over extended intervals:

$$\delta_{\text{eff}}(t) = \int_0^t \delta(\tau) \cdot w(\tau) d\tau$$

where  $w(\tau)$  is a persistence weighting function reflecting reduced corrective attenuation.

Subjective intensity  $I(t)$  is then hypothesized to depend on this effective discrepancy signal:

$$I(t) \propto \delta_{\text{eff}}(t)$$

Under high evaluative gain ( $g \geq g^*$ ),  $\delta_{\text{eff}}(t)$  remains low due to rapid correction, resulting in limited experiential impact. Under low gain ( $g < g^*$ ), persistent discrepancies contribute directly to subjective intensity.

Importantly, evaluative monitoring itself feeds back on the gain parameter. The act of evaluating, measuring, or attempting to control the state increases  $g$ :

$$g = g_0 + \alpha \cdot E(t)$$

where  $E(t)$  represents evaluative load and  $\alpha > 0$ . This introduces an intrinsic instability: as soon as the system attempts to monitor or stabilize the regime,  $g$  increases above threshold, restoring the standard coupling and collapsing the state.

This formulation captures three core properties of the proposed regime:

First, threshold dynamics: small changes in evaluative gain around  $g^*$  produce discontinuous changes in system behavior.

Second, persistence of discrepancy: prediction error is no longer transient but becomes a sustained signal contributing to experience.

Third, self-terminating structure: the same processes that would allow intentional control also reintroduce coupling, making the regime inherently unstable under monitoring.

This minimal model does not specify neural implementation but is compatible with architectures in which discrepancy signals (e.g., prediction error) and control signals (e.g., evaluative or salience-

related processes) are separable and dynamically coupled.

### **Why This Has Been Missed**

Despite the apparent familiarity of the underlying experiences, the regime described here has remained largely uncharacterized in formal accounts of positive affect. This absence is not necessarily due to rarity of the phenomenon, but to systematic features of how affect is studied, conceptualized, and operationalized.

A primary factor is methodological. Standard experimental approaches rely heavily on real-time measurement, explicit reporting, and task-based evaluation. These procedures implicitly introduce evaluative demands that increase control engagement. If access to the regime depends on reduced evaluative coupling, then such methods are not neutral observations but active interventions that suppress the phenomenon. As a result, the relevant states may be selectively absent from the very datasets used to build theoretical models.

A second factor is conceptual. Dominant frameworks treat positive affect as an output variable, typically scaled by reward magnitude, goal attainment, or appraisal. Within this view, affect is something to be produced, modulated, and optimized. The possibility that certain forms of positive affect emerge only when optimization processes are attenuated is therefore structurally excluded. This creates a blind spot: phenomena that are incompatible with optimization are either reinterpreted within existing models or dismissed as noise, anomaly, or anecdotal variation.

A third factor concerns classification pressure. When confronted with unusual or intense experiential reports, there is a tendency to map them onto established categories such as flow, meditative absorption, or nostalgia. While such mappings provide immediate interpretability, they may obscure mechanistic differences. In particular, similarities at the level of phenomenology can mask divergences in access conditions, control dynamics, and failure modes.

A fourth factor is reproducibility bias. Scientific practice favors phenomena that can be reliably induced, stabilized, and measured across repeated trials. By contrast, the regime described here is characterized by limited trainability, sensitivity to context, and instability under control. These properties make it difficult to capture using standard protocols, and therefore less likely to be incorporated into cumulative research programs.

Finally, there is a structural asymmetry in what is considered theoretically tractable. Processes that involve active control, optimization, and measurable outputs are more easily formalized and integrated into existing frameworks. Regimes defined by the attenuation of control, especially when they resist direct manipulation, fall outside this tractable space. As a result, they remain underrepresented despite their potential relevance.

Taken together, these factors suggest that the absence of such regimes from dominant theories does not necessarily reflect their insignificance, but rather the interaction between methodological constraints and conceptual assumptions. By explicitly accounting for these constraints, the present framework aims to make these regimes accessible to systematic investigation.

## **Conclusion**

The present paper proposes that high-intensity positive affect is not adequately captured by models that treat it as a direct outcome of reward, learning, or goal-directed optimization. Instead, joy is conceptualized as a regime-dependent property of the system, emerging when evaluative coupling falls below a critical threshold and discrepancies remain informational rather than being rapidly converted into corrective demands.

This framework accounts for a set of observations that are difficult to reconcile with existing approaches, including abrupt and all-or-none transitions, sensitivity to evaluative context, limited trainability, and the paradoxical effects of measurement. It also provides a unifying interpretation of phenomena spanning development, memory reactivation, perceptual engagement, and pharmacological dissociations, by linking them to a common control parameter governing the relationship between discrepancy detection and evaluative correction.

By introducing a regime-based perspective, the model shifts the focus from the production of positive affect to the conditions that enable its accessibility. In this view, the central question is not how to increase reward or optimize behavior, but how to modulate the structure of control such that discrepancies can persist and propagate within the system.

Importantly, the framework is designed to be empirically vulnerable. It generates specific predictions that distinguish it from gradualist and reward-based accounts, particularly regarding discontinuity of onset, disruption by evaluation, and the absence of reliable training effects. These predictions provide a basis for direct experimental tests, as well as for methodological revisions that minimize interference with the phenomena under study.

More broadly, the proposal highlights a potential blind spot in affect research: phenomena that depend on reduced evaluative control may be systematically underrepresented due to the very methods used to investigate them. Addressing this limitation may require not only new models, but new experimental strategies that treat observation itself as a variable rather than a neutral process.

Taken together, these considerations suggest that joy, in at least one of its forms, is not best understood as an output to be maximized, but as a regime to which access must be permitted.

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